6.7.5 Point of intersection of two straight lines

If $\frac{x-x_1}{a_1}=\frac{y-y_1}{a_2}=\frac{z-z_1}{a_3}$ and $\frac{x-x_2}{b_1}=\frac{y-y_2}{b_2}=\frac{z-z_2}{b_3}$ are two lines, then every point on the line is of the form $(x_1+sa_1, y_1+sa_2, z_1+sa_3)$ and $(x_2+tb_1, y_2+tb_2, z_2+tb_3)$ respectively. If the lines are intersecting, then there must be a common point. So, at the point of intersection, for some values of s and t, we have

$$(x_1+sa_1, y_1+sa_2, z_1+sa_3) = (x_2+tb_1, y_2+tb_2, z_2+tb_3)$$

$$\text{Therefore, } x_1+sa_1 = x_2+tb_1, y_1+sa_2 = y_2+tb_2, z_1+sa_3 = z_2+tb_3$$

By solving any two of the above three equations, we obtain the values of s and t. If s and t satisfy the remaining equation, the lines are intersecting lines. Otherwise the lines are non-intersecting . Substituting the value of s, (or by substituting the value of t), we get the point of intersection of two lines.

If the equations of straight lines are given in vector form, write them in cartesian form and proceed as above to find the point of intersection.

Example 6.33

Find the point of intersection of the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{5}=\frac{y-1}{2}=z$.

Solution

Every point on the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}=s$ (say) is of the form $(2s+1, 3s+2, 4s+3)$ and every point on the line $\frac{x-4}{5}=\frac{y-1}{2}=z=t$ (say) is of the form $(5t+4, 2t+1, t)$. So, at the point of intersection, for some values of s and t, we have

$$(2s+1, 3s+2, 4s+3) = (5t+4, 2t+1, t)$$

Therefore, $2s-5t=3$, $3s-2t=-1$ and $4s-t=-3$. Solving the first two equations we get $t=-1$, $s=-1$. These values of s and t satisfy the third equation. Therefore, the given lines intersect. Substituting, these values of t or s in the respective points, the point of intersection is $(-1,-1,-1)$.

6.7.6 Shortest distance between two straight lines

We have just explained how the point of intersection of two lines are found and we have also studied how to determine whether the given two lines are parallel or not.

Definition 6.6

Two lines are said to be **coplanar** if they lie in the same plane.

Note

If two lines are either parallel or intersecting, then they are coplanar.

Definition 6.7

Two lines in space are called **skew lines** if they are not parallel and do not intersect

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